

IT ARCHITECTS CREDIT CARD FRAUD DETECTION

TEAM MEMBERS



Mr Arth Patel

Role : Front-End Developer



Mr Tyler Helmirch

Role: Scrum Master



Mr Jay Patel

Role: Back-End Developer



Mr Xavier Jackson

Role: Machine Learning Lead



Mr Zenil Moradia

Role: Quality assurance

PROBLEM STATEMENT

Credit card fraud remains the leading form of identity theft in the United States, with 2024 seeing over 449,000 complaints to the Federal Trade Commission and within the first quarter of 2025 adding another 151,000 cases. In the United States, 63% of cardholders have been targeted at least once, and more than half of these victims report experiencing repeated attacks.

This persistent threat not only inflicts hundreds of millions of dollars in annual losses but also erodes consumer confidence in digital payments. Existing detection systems struggle to keep up with the evolving tactics of scammers and the extreme imbalance between legitimate and fraudulent transactions. There is an urgent need for a robust, real-time solution that can accurately flag while minimizing false alarms.

PROBLEM DESCRIPTION

Project Name:	Machine Learning Model for Credit Card Fraud Detection
Team:	IT Architects
Project Description:	Current deterrents include rule-based fraud detection techniques, whereas our goal is to implement an AI-powered fraud-detection service that catches up to 80% of fraud attempts in real time while cutting false positives by 30%, so issuers lose less money and cardholders enjoy seamless, secure spending. This project aims to deliver an advanced machine-learning classifier for credit-card fraud detection, designed for issuers and cardholders frustrated by today's rule-based systems and high false-alarm rates. Leveraging SMOTE to rebalance data and XGBoost for precision, we'll identify fraudulent transactions before funds leave the account, reduce false positives on legitimate purchases, and trigger real-time alerts that let cardholders act immediately. Our mission is to replace rigid, hand-tuned rules with a data-driven model that adapts to evolving fraud patterns by minimizing losses, preserving reputations, and restoring trust in digital payments.
Benefit Outcomes:	Credit Card companies can expect a reduction in false alarms, monetary losses, and damage to their reputation. Credit Card holders can expect a reduction in unauthorized charges, fewer declined legitimate transactions, and greater peace of mind when using their cards as well as an increase in user satisfaction and engagement.
Github Link:	https://github.com/htmw/F2025-Team I/wiki

PERSONA I: SARAH MILLER

- Age: 28
- Occupation: Graphic Designer
- Sarah is an avid online shopper who loves finding unique items from small businesses. She uses her credit card for most of her purchases, appreciating the convenience and rewards. She's tech-savvy and expects seamless digital experiences. Recently, her card details were stolen from a third-party website, leading to several unauthorized small purchases. The process of getting a new card and updating her payment information on various subscriptions was a significant hassle and made her wary of future online transactions. She's looking for reassurance and security in her digital financial interactions.
- Goals:
 - Wants to shop online without constant worry about card fraud.
 - Needs quick and clear notifications for any suspicious activity on her accounts.
 - Hopes for an easy way to verify transactions or freeze her card directly from her phone if needed.
- Frustrations:
 - The time and effort involved in cancelling cards and updating payment details after fraud.
 - Feeling vulnerable and exposed when her financial information is compromised.
 - Receiving generic, unhelpful alerts that don't provide clear next steps.
- Needs from the Solution:
 - Real-time alerts that are easy to understand and act upon.
 - An intuitive mobile interface that allows her to quickly approve or deny a transaction and temporarily lock her card.
 - Minimal false positives on her legitimate purchases so she doesn't lose trust in the system.

PERSONA 2: ALEX CHEN

- Age: 32
- Occupation: Founder of "PetPals Boutique" | The E-commerce Startup Founder
- Alex recently launched "PetPals Boutique," an online store selling custom pet accessories. Business is booming, but he's starting to see a disturbing trend of chargebacks due to fraudulent orders. These chargebacks not only cost him money but also damage his store's reputation with payment processors. He's passionate about his business but has limited resources and can't afford a dedicated fraud team. He needs an automated, intelligent system that can protect his revenue and allow him to focus on growing PetPals Boutique, not fighting fraud.
- Goals:
 - To minimize chargebacks and financial losses from fraudulent transactions.
 - To protect his business's reputation with payment providers.
 - To have a simple, affordable, and effective fraud detection solution that integrates easily with his e-commerce platform.
- Frustrations:
 - The financial impact of fraudulent orders and chargeback fees.
 - The time-consuming process of manually reviewing suspicious orders.
 - Fear that his small business could be targeted by organized fraud rings, which he feels ill-equipped to handle.
- Needs from the Solution:
 - An easy-to-integrate API that works seamlessly with his e-commerce platform.
 - A dashboard that provides clear, actionable insights on potential fraud, without requiring a lot of technical expertise.
 - Automated decision-making that blocks high-risk transactions instantly and allows low-risk ones to proceed without delay.

PERSONA 3: MARCUS JOHNSON

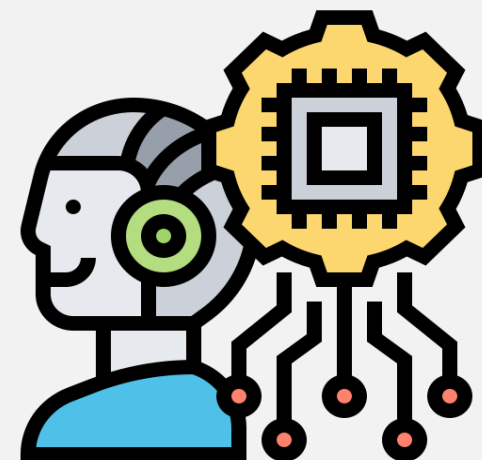
- Age: 49
- Occupation: Head of Fraud Prevention at "SecureBank" | The Bank Security Officer
- Marcus leads the fraud prevention department at a mid-sized bank. His team handles thousands of credit card transactions daily, constantly battling sophisticated fraud schemes. With new tactics emerging all the time, his biggest challenge is staying ahead of the criminals while ensuring legitimate customer transactions are processed smoothly. He's looking for a powerful, scalable solution that can integrate with their existing systems, reduce false positives, and provide actionable intelligence to his team. He needs to improve detection rates, reduce operational costs associated with fraud investigation, and enhance customer trust.
- Goals:
 - To significantly reduce the bank's annual losses from credit card fraud.
 - To decrease the rate of false positives, improving customer experience and reducing operational overhead.
 - To implement a system that provides real-time, adaptive fraud detection capable of learning new patterns.
- Frustrations:
 - The constant arms race with fraudsters and the difficulty of keeping up with evolving tactics.
 - The pressure to maintain a high level of security without inconveniencing legitimate customers.
 - The sheer volume of data and the challenge of sifting through it to find genuine fraud.
- Needs from the Solution:
 - A highly scalable system that can process millions of transactions in real time.
 - Advanced machine learning capabilities that can adapt to new fraud patterns and minimize false positives.
 - Customizable rules and thresholds that allow his team to fine-tune the system's behavior.

PRODUCT PROTOTYPE



TECHNOLOGY AND TOOLS

- Data Processing and Feature Engineering
 - **Pandas / NumPy** – Core tabular and numerical routines.
 - **Dask** – Scales Pandas workflows to larger-than-memory datasets with minimal code changes.
 - **Imbalanced-learn** – SMOTE, ADASYN, and other resampling techniques built atop scikit-learn APIs.
- Model Development and Experimentation
 - **Scikit-learn** – Standardized pipeline, preprocessing, and model-selection tools.
 - **XGBoost / LightGBM / CatBoost** – Gradient boosting implementations with GPU support for speed and advanced class-imbalance handling.
 - **Optuna / Hyperopt** – Efficient hyperparameter optimization via Bayesian search or Tree-structured Parzen Estimator.
 - **MLflow** – Track experiments, log parameters/metrics, and register model versions in one place.
- Visualization and End-User Interface
 - **Plotly / Dash** – Interactive EDA dashboards you can hook up to live model inference.
 - **Streamlit** – Superfast prototyping of apps—forms for inputting mock transactions, real-time results.
 - **Tableau / Power BI** – Drag-and-drop business dashboards for non-technical stakeholders, integrating CSV or database outputs.





TECHNOLOGIES CONT.

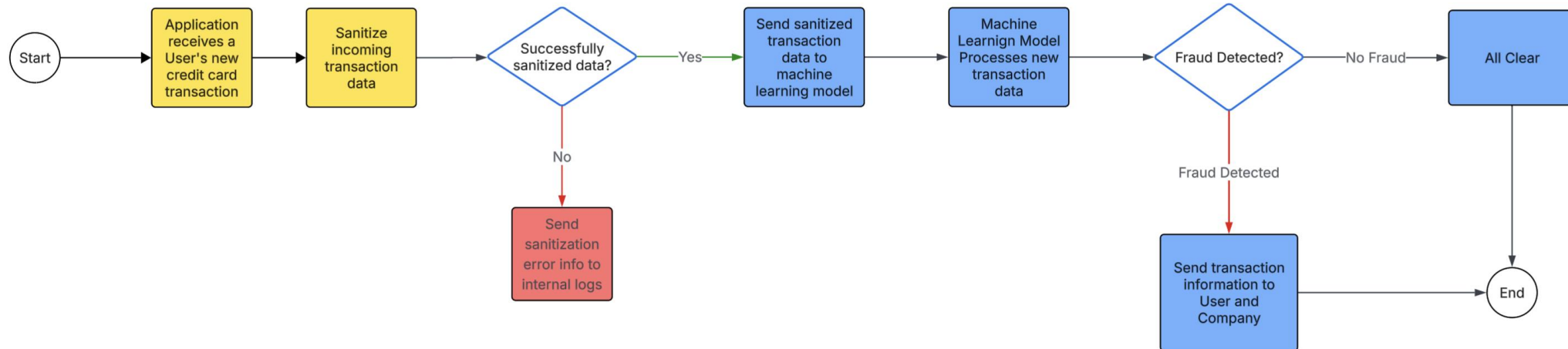
- Deployment
 - **FastAPI** – High-performance REST endpoints for batch or real-time scoring.
 - **Docker** – Containerizes entire pipeline (preprocessing → model → API) for consistency.
 - **Kubernetes** – Autoscale inference pods behind a load balancer; keeps service resilient.
 - **Terraform** – Infrastructure-as-code to spin up cloud resources reproducibly (EKS/GKE/AKS, databases, VPCs).
- Collaboration
 - **GitHub / GitLab** – Version control for code, issue tracking, and CI/CD integration with Actions or Runners.
 - **WikiPages**
- Data Storage & Logging
 - MongoDB



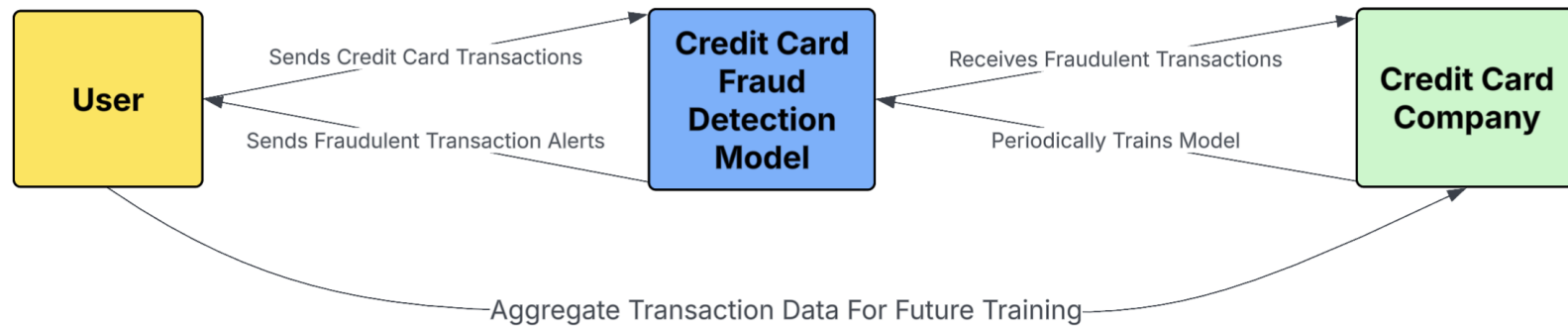
DIFFERENT MODELS

Component	Explanation	Project Goal Alignment
The Problem (Imbalance)	If we trained a model on the raw data, it would achieve 99.83% accuracy just by guessing "legitimate" every time, making it useless for catching actual fraud.	Justifies why your model is necessary to meet the 80% fraud capture goal.
The Solution: SMOTE	We use SMOTE (Synthetic Minority Over-sampling Technique) to mathematically generate new, synthetic fraud examples that are similar to the real ones. We only do this on the training data to balance the classes and force the model to learn the fraud patterns effectively.	Addresses the severe class imbalance and improves the model's ability to achieve high Recall (catching fraud).
The Classifier: XGBoost	We use XGBoost, a powerful gradient boosting algorithm. It's excellent for tabular data and has advanced internal mechanisms (like using the scale_pos_weight parameter) to handle imbalanced datasets better than simpler models.	This choice is key to minimizing losses (high Recall) and achieving high Precision (cutting false positives).
The Output (Real-Time)	The trained model is deployed via a high-speed FastAPI service to provide predictions in milliseconds. The output is not just a score, but an Actionable Alert (e.g., "Transaction Blocked" or "Approve/Deny") displayed in the Streamlit front-end.	This satisfies the core need of your personas, especially Sarah Miller (quick alerts) and Alex Chen (real-time protection).

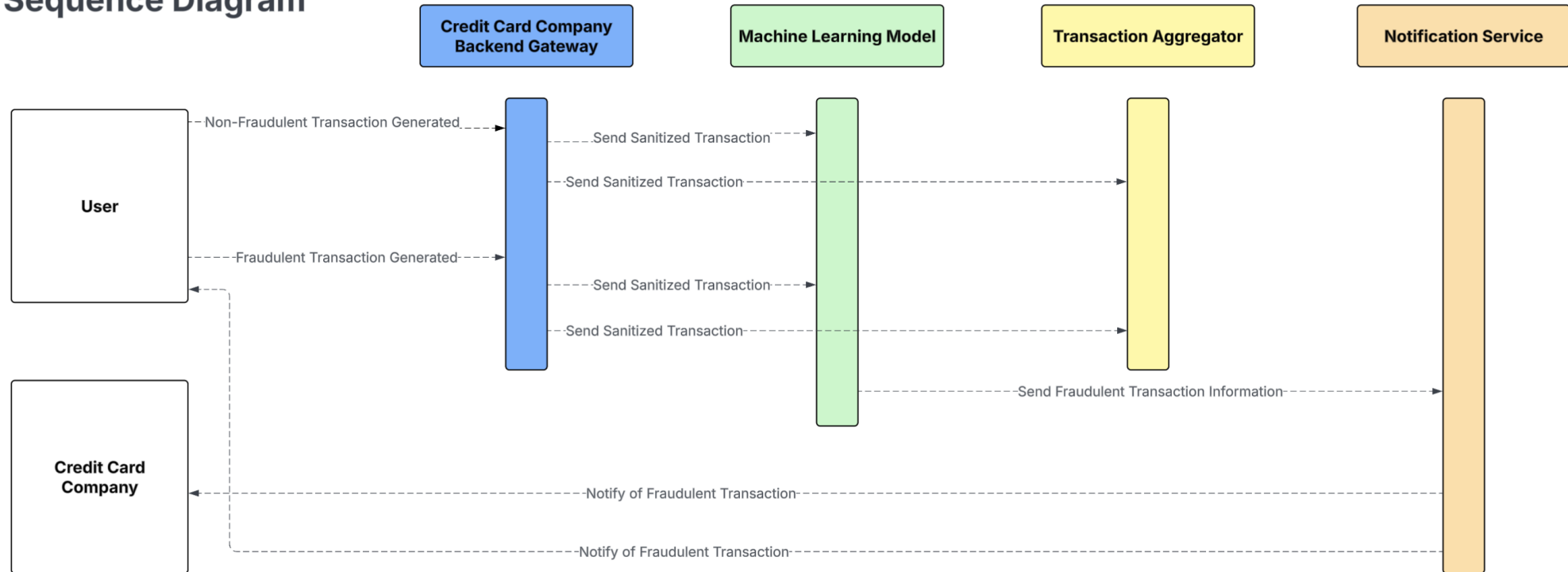
Credit Card Fraud Model State Diagram



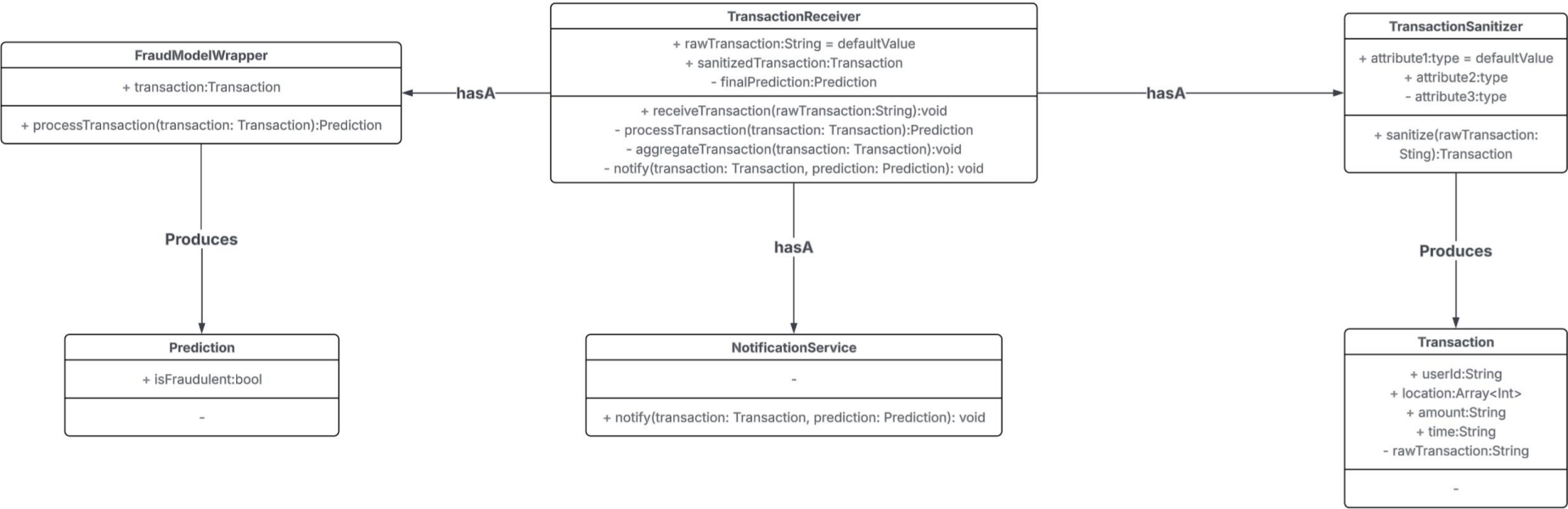
Credit Card Fraud Model Context Diagram



Credit Card Fraud Model Sequence Diagram



Credit Card Fraud Model Class Diagram



USER STORIES

User Story	Acceptance Criteria	Feature		
As a user, I want the system to automatically receive and log new credit card transactions.	The app receives transactions in real time; Each transaction is timestamped and stored.	Transaction Receiver		
As a user, I want invalid or incomplete transactions to be rejected before analysis.	Transactions missing fields like amount or merchant ID are dropped; Dropped transactions are logged with a reason.	Transaction Sanitizer		
As a user, I want all transactions to be formatted consistently before being processed by the model.	Currency, timestamp, and amount fields are standardized; Cleaned transactions are passed to the model.	Transaction Sanitizer		
As a user, I want each transaction to be scored for fraud likelihood.	Model returns a fraud probability score between 0 and 1; Transactions with missing features are skipped and logged.	Fraud Detection Model Wrapper		
As a user, I want the app to classify transactions as fraudulent or safe based on a configurable threshold.	The threshold value can be set manually; Transactions above threshold are marked as 'Fraudulent'.	Fraud Detection Model Wrapper		
As a user, I want to be notified immediately when a fraudulent transaction is detected.	Notification includes merchant, amount, and time; Notification is delivered within 5 seconds of detection.	Notification Service		
As a user, I want legitimate transactions to be logged for auditing purposes.	Non-fraudulent transactions are recorded with timestamp and merchant; No notification is sent.	Transaction Receiver		
As a user, I want duplicate transactions to be ignored.	Duplicate detection based on transaction ID; Only first instance is processed; Duplicates logged.	Transaction Sanitizer		
As a user, I want to mark a transaction as 'Not Fraudulent' if it was wrongly flagged.	User can flag false positives through the app; Feedback is stored for model retraining.	Notification Service		
As a data scientist, I want to export flagged transactions for model retraining.	Fraudulent and legitimate labels are stored; Export available as JSON or CSV.	Fraud Detection Model Wrapper		
As a user, I want to choose how to receive alerts (SMS, push, or email).	User selects preferred channel; Notifications are routed correctly.	Notification Service		
As a user, I want to view a list of past transactions and their fraud status.	Transactions can be filtered by date or status; History includes fraud score and decision.	Notification Service		
As a developer, I want a way to check if all system components are running.	A /health endpoint returns component statuses; Returns 200 OK only when all services are operational.	Transaction Receiver		
As a developer, I want to track latency between transaction receipt and notification.	Start and end timestamps recorded for every transaction; Average latency reported in logs.	Transaction Receiver		
As a user, I want to see a summary report of detected fraud over time.	Report shows counts by day/week/month; Fraud rate percentage is calculated.	Notification Service		

SPRINT-I TEST CASES

Test Case ID	Scenario Description	Input Data	Expected Output	Category
TC001	Normal low-value transaction	\$50, New York, Card used before	Legit	Normal Transaction
TC002	High-value transaction in abroad nation	\$5,000, Germany, User location: USA	Fraud	Location Anomaly
TC003	Multiple transactions in short time	10 transactions in 4 minutes, total \$5000	Fraud	Velocity Check
TC004	Online purchase from familiar vendors	\$200, amazon	Legit	Trusted Vendor
TC005	Transaction at unusual hour	\$600, 3:00AM, Normal hours 9 AM, 9 PM	Suspicious	Time-based Anomaly

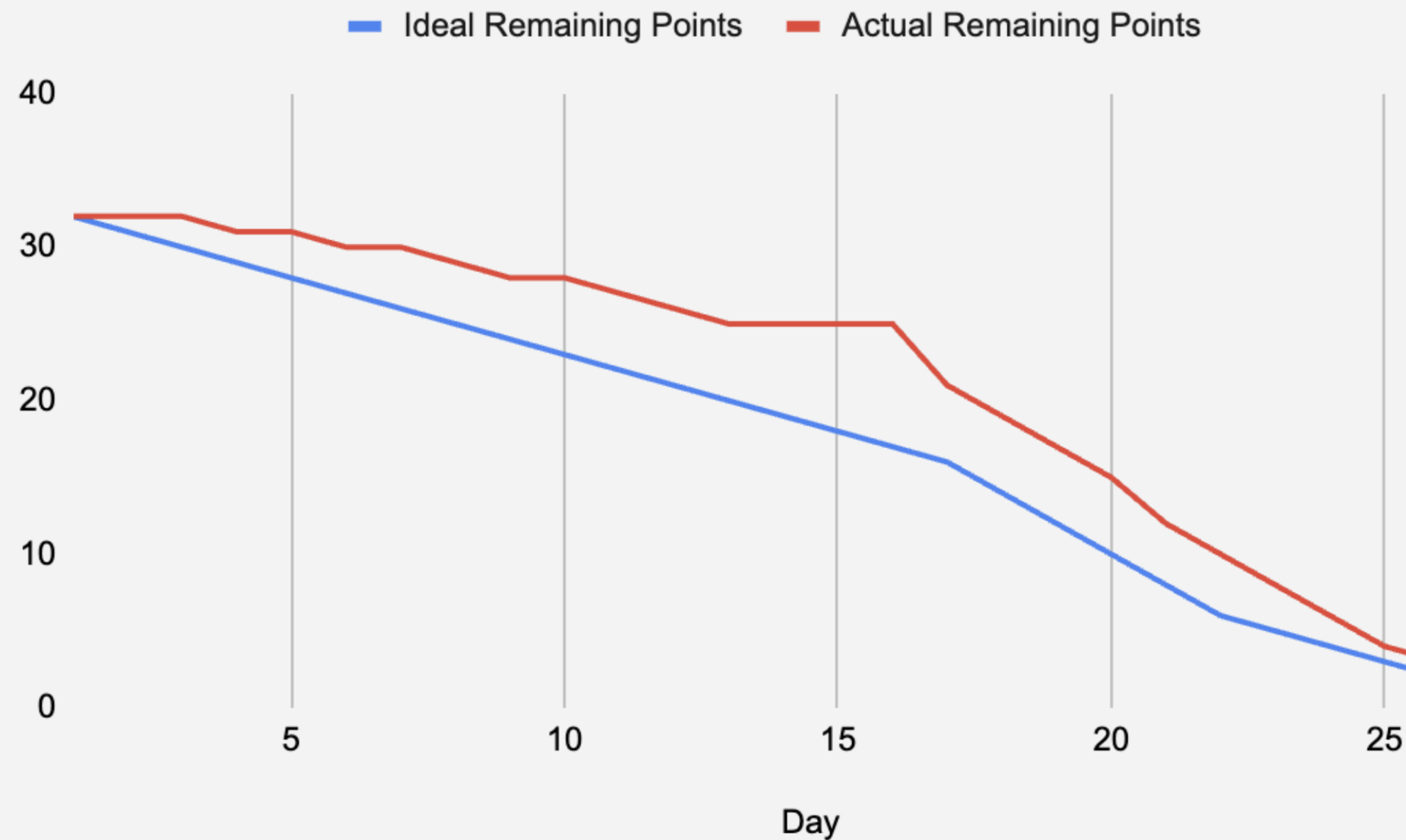
SPRINT-I TEST CASES

TC006	Card used after being reported lost	\$500, Card status: Reported lost	Fraud	Stolen Card
TC007	Mismatch in billing and shipping country	Billing: USA, Shipping: Australia	Fraud	Geographical Mismatch
TC008	Multiple declines followed by approval	3 declines, 1 approval for \$900	Fraud	Decline Pattern
TC009	Small test transaction followed by large one	\$1 then \$2,000 within 5 mins	Fraud	Test Transaction Pattern
TC010	Purchase with consistent spending history	\$70, Groceries	Legit	Behavioral Match

SPRINT-I TEST CASES

TC011	Transaction from new device/IP	Device: Unknown, IP: new region	Suspicious	Device Anomaly
TC012	Cross-border ATM withdrawal	Japan, User location: USA	Fraud	Geo Velocity
TC013	Sudden change in spending category	\$5000, Category: Luxury Jewelry	Suspicious	Behavioral Deviation
TC014	Repeated identical transactions	5 identical \$500 transactions in 20 mins	Fraud	Duplication Pattern
TC015	Refund request immediately after purchase	\$1500 refunded after 10 mins	Suspicious	Refund Abuse

Sprint 1 Burndown Chart



SPRINT RETROSPECTIVE

- What we did well: Good sprint planning and collaboration
- What we felt didn't go well: There were stretches of the sprint where we weren't able to meet and had to rush more at the end of the sprint as reflected in the burndown chart
- What we want to do differently next time: Daily standups were difficult because we all work and have school, want to start doing asynchronous status updates over WhatsApp